

Interpretable Baseline Pipeline for Corrosion Image Analysis, Hidden Damage Estimation, and Exploratory Proxy-RUL

Scientific report based on the implemented codebase, generated outputs, diagnostics, and benchmark analysis

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Abstract

This report summarizes the current state of an interpretable, leakage-safe baseline pipeline for corrosion image analysis in reinforced specimens observed longitudinally over time. The project uses a staged formulation — surface corrosion progression → hidden damage estimation → degradation modelling → proxy-RUL — because the dataset does not support direct supervised remaining useful life prediction. After audit and alignment, the usable dataset contains 791 image-table observations from 48 specimens, with dense surface labels but only 48 terminal structural-label rows. The strongest parts of the project are the data audit, deterministic specimen mapping, interpretable image-feature extraction, temporal corrosion progression analysis, feature diagnostics, and robustness-oriented benchmark review. The surface stage mainly validates the image-analysis pipeline, especially through the near-identity between the extracted rust-area feature and the labeled surface rust percentage. The structural stage remains weak and confounded, with performance deteriorating sharply under leave-one-campaign-out evaluation and the most robust final models becoming metadata-driven rather than image-driven. The degradation and proxy-RUL stages remain exploratory and should be interpreted as descriptive downstream analyses rather than validated prognostics. The report highlights the figures and findings that are most suitable for presentation while explicitly stating the current limitations and the next scientifically defensible step.

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1 Introduction

Corrosion monitoring is important for structural health monitoring because visible surface degradation may precede or accompany deeper internal damage. Image-based monitoring is attractive because it is scalable and relatively easy to collect, but converting visible corrosion patterns into defensible structural conclusions is difficult. This project addresses that challenge through an interpretable baseline rather than a black-box end-to-end model.

The project should be understood as a careful attempt to answer a staged scientific question:

surface corrosion progression → hidden damage estimation → degradation modelling →
proxy-RUL

That wording matters. The codebase does not claim that the dataset supports direct image-to-RUL learning. Instead, it builds and evaluates each stage separately, adds diagnostics to reveal where the evidence is strong or weak, and uses grouped validation to avoid specimen leakage.

2 Project Objective and Scientific Motivation

The main objective is to build a scientifically sound baseline pipeline that can:

- quantify visible corrosion from images using interpretable features,
- test whether those features help estimate hidden structural damage,
- summarize specimen-level degradation behaviour over time, and
- generate an exploratory proxy-RUL view through threshold-status analysis.

The motivation for this staged design is practical and scientific:

- the dataset contains many repeated image observations,
- visible corrosion information is available on all rows,
- structural labels exist only on a very small terminal subset, and
- robust, explainable baseline evidence is needed before stronger modelling claims can be justified.

The solution therefore prioritizes interpretability, leakage safety, and benchmark robustness over flashy modelling complexity.

3 Dataset and Documentation

3.1 Data assets and verified facts

The raw assets are:

- workbook: `Data/Images_Dataset_A-Z.xlsx`,
- image folder: `Data/Images_dataset/`,

- thesis PDF: `Documentation/Thesis/...`,
- conference paper PDF: `Documentation/Conferences/....`

After audit and alignment, the canonical dataset facts are shown in Table 1.

Table 1: Verified dataset facts after audit and alignment.

Item	Value
Image files found	792
Usable aligned observations	791
Unique specimens	48
Grouped split key	<code>specimen_id</code>
Rows with structural labels	48
Structural-label fraction	$48 / 791 = 6.1\%$
Corrupted or orphan image	<code>E01-20240508-17W.png</code>
Campaign 1 duration	week 0 to week 28
Campaign 2 duration	week 0 to week 36

3.2 Longitudinal and grouped structure

This is a repeated-observation dataset. Each specimen appears at multiple time points, so rows are not independent. That has two direct consequences:

- the same specimen must never appear in both train and test sets,
- progression over time should be analysed at both row level and specimen level.

The grouped nature of the data is therefore central to the whole modelling strategy.

3.3 Dense surface labels and sparse structural labels

Surface labels are available on all usable rows, especially:

- `surface_total_rust_pct`,
- `peak_rust_pct`,
- categorical bins of the same visible corrosion quantities.

Structural labels are sparse and terminal-only:

- `wire_area_loss_frac`,
- `ultimate_load_kn`.

Each specimen has exactly one structural-label row, and those rows occur at the final observation time. This is the main scientific reason the project does not attempt direct RUL supervision.

3.4 What the thesis and documentation contributed

The thesis and conference paper were used to reconstruct scientific context that is not explicit in the workbook alone. In particular, they contributed:

- specimen naming and campaign logic,
- treatment-group interpretation,
- the meaning of the structural targets,
- the strip-wise corrosion analysis concept,
- the approximate RGB threshold logic for rust, black, and gray regions.

An important limitation remains: the exact original GIMP preprocessing pipeline could not be fully recovered, so the implemented image-feature stage uses the provided PNGs directly and applies deterministic feature extraction.

4 Problem Formulation

4.1 Why direct supervised RUL is not valid

Direct supervised RUL prediction would require image-level or time-level residual-life labels across the observation window. This dataset does not provide that. Instead, it provides:

- dense visible corrosion measurements at many time points,
- sparse structural measurements only at terminal rows,
- no direct failure time label attached to each image.

Therefore, the scientifically valid formulation is:

visible corrosion progression → hidden damage estimation → degradation modelling →
proxy-RUL

4.2 Why grouped splitting is necessary

Because each specimen appears repeatedly, ordinary random splitting would create leakage. The implemented split strategies are therefore specimen-safe:

- repeated grouped holdout by specimen,
- leave-one-treatment-out,
- leave-one-campaign-out.

Among these, leave-one-campaign-out is the hardest and most scientifically informative stress test because it tests whether the learned pattern survives a campaign shift rather than merely reusing specimen- or campaign-specific shortcuts.

4.3 Why interpretable image features were used first

The first image stage is intentionally interpretable. Instead of training a deep end-to-end model immediately, the project starts from feature families that can be described physically:

- rust-mask area,
- brightness and color statistics,
- texture descriptors,
- morphology of rust regions,
- strip-wise corrosion summaries along the specimen length.

This choice makes the pipeline auditable and makes it easier to discover when a benchmark is genuinely informative versus when it is mainly reconstructing a precomputed visible corrosion quantity.

5 Implemented Pipeline

5.1 Data loading and audit

The pipeline begins by loading the workbook and scanning the image directory. The code validates:

- required columns,
- sample-name parsing,
- consistency between specimen IDs and filename-derived IDs,
- image readability,
- workbook-image alignment,
- uniqueness of observation identifiers.

This stage produces the canonical aligned table `outputs/data/master_table.csv` and logs issues explicitly rather than discarding them silently.

5.2 Preprocessing and validation

A deterministic mapping file, `configs/specimen_mapping.yaml`, reconstructs campaign and treatment structure. This matters because the workbook treatment columns are too coarse to identify the scientific treatment protocols reliably on their own.

The audit also resolves an important unit issue: the workbook column for wire area loss is labeled as a percentage but is numerically stored as a fraction. Internally, the project treats it as `wire_area_loss_frac` and derives a percentage version explicitly where needed.

5.3 Image feature extraction

The image-feature stage applies thesis-aligned threshold logic and deterministic feature extraction. The main output is `outputs/features/image_features.csv`. The extracted features fall into five broad families:

- mask features,
- color and brightness features,
- histogram and texture features,
- morphology features,
- spatial strip features.

This stage is scientifically useful because it exposes which visible corrosion characteristics are strongly measurable from images and which engineered features are unstable or redundant.

5.4 Feature engineering

The metadata and image features are merged into `outputs/data/full_feature_table.csv`. Stage-specific feature filtering is then applied.

The main fixed modelling exclusions are:

- `img_strip_count`, which is constant under the current strip geometry,
- `img_contrast`, which duplicates `img_brightness_std`,
- `img_edge_to_center_rust_ratio`, which is unstable in low-rust images.

5.5 Split strategy

The same grouped split logic is reused across modelling stages. Split manifests are saved to disk, and the summary files confirm that no specimen leakage was detected in the generated splits.

5.6 Surface modelling

The surface stage benchmarks interpretable features against two visible corrosion targets:

- `surface_total_rust_pct`,
- `peak_rust_pct`.

This stage is useful, but it must be interpreted carefully. The first target is largely a label-reconstruction sanity check because it is almost numerically identical to the extracted rust-mask area ratio. The second target, `peak_rust_pct`, is a more meaningful auxiliary surface benchmark because it is not an exact copy of a single feature, even though it is still strongly related to the strip-based feature family.

5.7 Hidden-damage modelling

The hidden-damage stage uses the terminal structural subset. Candidate models are:

- Random Forest,
- Gradient Boosting,
- XGBoost,
- CatBoost.

The improvement round added:

- feature-space cleanup,
- feature-family ablations,
- robustness-aware model selection across split strategies,
- before/after comparison against the earlier baseline.

The key outcome is important: the final robust selections became metadata-only models for both structural targets. That is useful scientifically because it shows where the current robust signal is coming from, but it also means the image features are not yet providing strong structural generalization.

5.8 Degradation modelling

The degradation stage uses the selected hidden-damage model to predict hidden damage across all observations. Those predictions are then:

- clipped to a physically plausible range,
- converted to a monotone proxy trajectory per specimen,
- fit with simple candidate families such as linear, log-time, logistic, Gompertz, and monotone isotonic regression.

This stage is best understood as a descriptive smoothing layer. It is useful for building specimen-level trajectories and threshold-status summaries, but it is not yet strong evidence of predictive degradation identification.

5.9 Proxy-RUL logic

The final stage evaluates whether the degradation trajectories cross hidden-damage thresholds of 0.20, 0.30, and 0.40 within a fixed projection horizon.

The key point is interpretive:

- the outputs are mainly threshold-status summaries,
- the current pipeline produces no forward threshold crossings within the projection horizon,
- therefore the stage remains exploratory rather than decision-ready.

6 Exploratory Analysis and Diagnostics

6.1 Dataset structure

The project added an explicit diagnostics layer because the raw pipeline outputs alone are not enough for presentation. The diagnostics turn important CSV outputs into readable figures and help answer the questions:

- what does the dataset actually look like,
- which features are useful or redundant,
- where do the models generalize,
- where do they fail.

6.2 Missingness

Figure 1 shows the overall missingness pattern. Figure 2 focuses specifically on the structural-label sparsity.

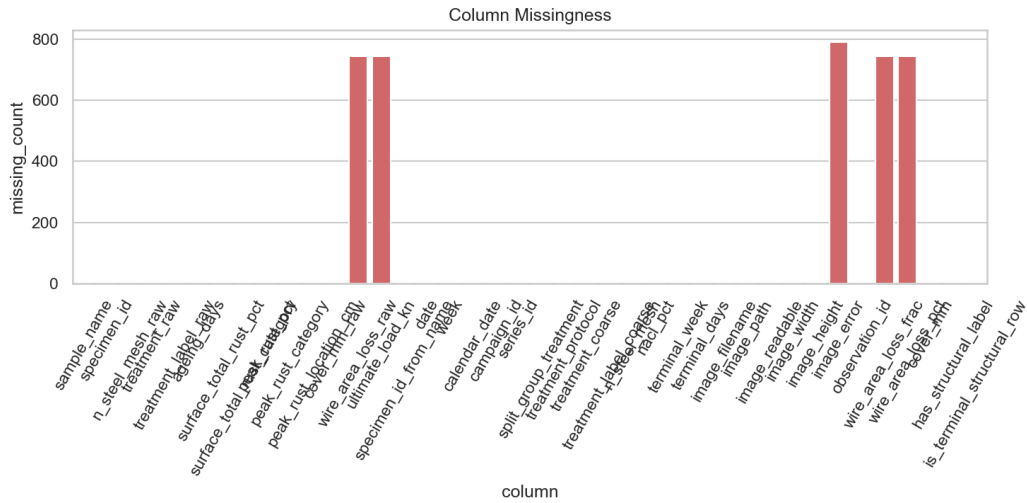


Figure 1: Overall missingness structure in the aligned dataset. Surface variables are dense, while structural variables are sparse and terminal-only.

Interpretation and evidence strength. Figure 1 is strong evidence because it directly shows that the dataset is rich for visible corrosion analysis but sparse for structural supervision. This is one of the clearest justifications for avoiding direct supervised RUL prediction.

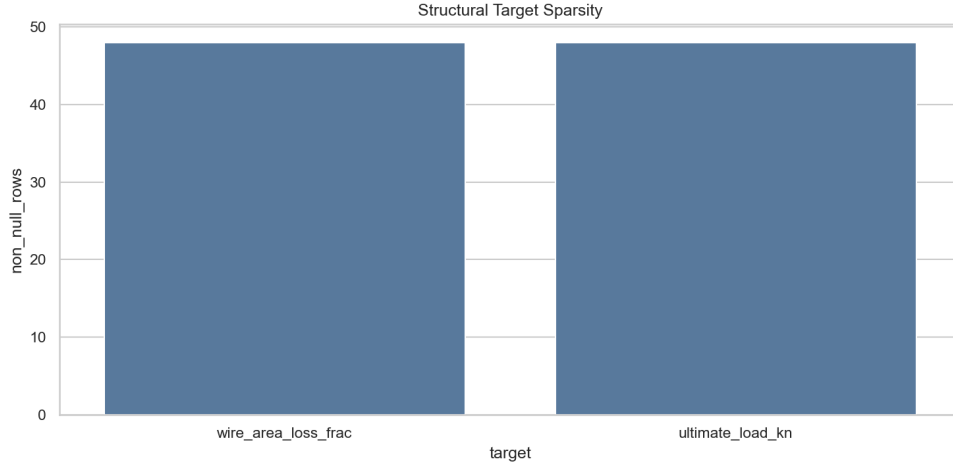


Figure 2: Structural-label sparsity. Only one structural row per specimen is available, and those rows occur at terminal time points.

Interpretation and evidence strength. Figure 2 is also strong evidence. It supports the report’s central formulation: visible corrosion can be studied densely, but hidden damage can only be studied through a small terminal subset.

6.3 Feature distributions

The feature-distribution diagnostics reveal a mixed picture:

- many visible corrosion features have sensible broad ranges,
- some engineered features are strongly skewed,
- some features are effectively degenerate.

6.4 Outliers

The project deliberately kept outliers visible and added readability-focused views rather than hiding them. Figure 3 shows selected skewed variables with plotting truncated at the 99th percentile for readability only.

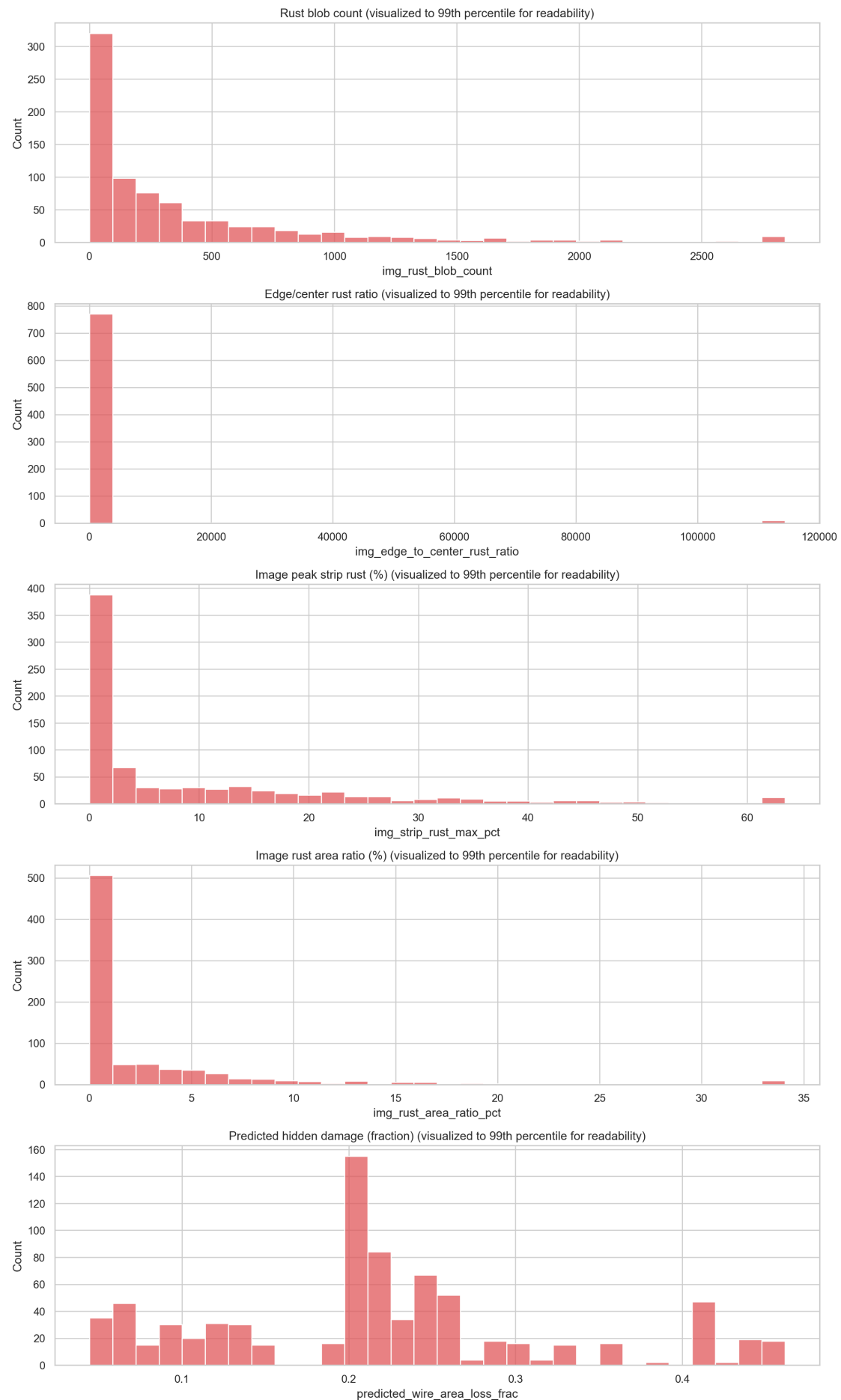


Figure 3: Readability-focused histograms for heavily skewed variables. The plotting range is truncated at the 99th percentile for visibility, while full-range statistics are kept separately in the diagnostics tables.

Interpretation and evidence strength. Figure 3 is strong evidence for the claim that the feature space contains heavy tails and unstable engineered variables. This is especially important for explaining why feature cleanup was not cosmetic.

6.5 Temporal progression

Visible corrosion progression is one of the most meaningful parts of the dataset. Figure 4 shows the campaign-level progression summary, while Figure 5 shows representative specimen trajectories.

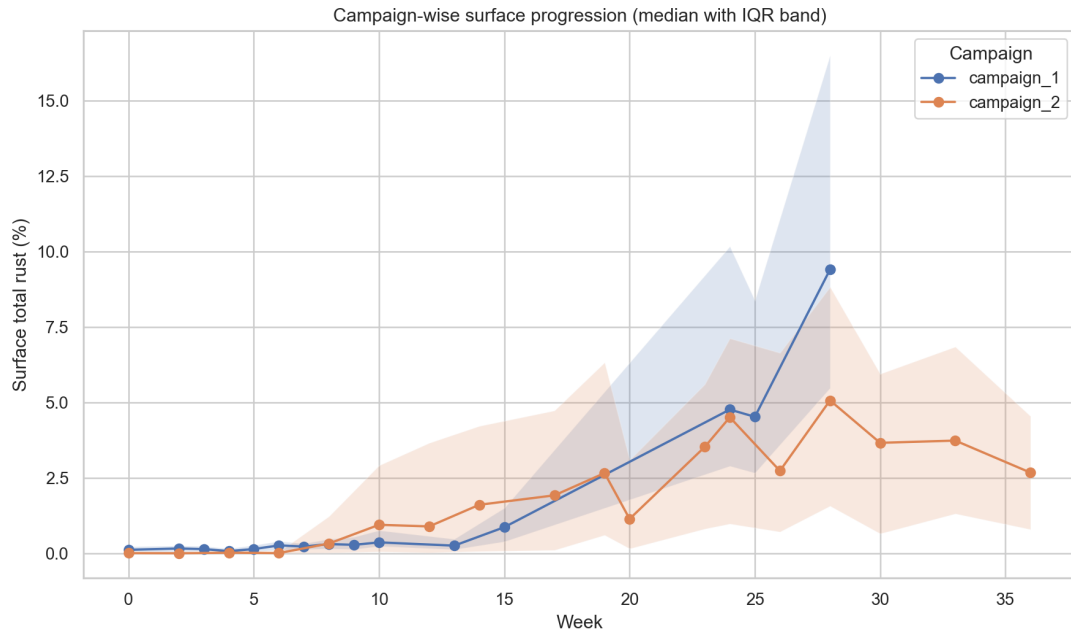


Figure 4: Campaign-level visible corrosion progression over time, shown as median behaviour with spread.

Interpretation and evidence strength. Figure 4 is strong evidence. It shows that visible corrosion generally increases with exposure time and that the two campaigns differ in both magnitude and spread. This is one of the best presentation figures in the project.

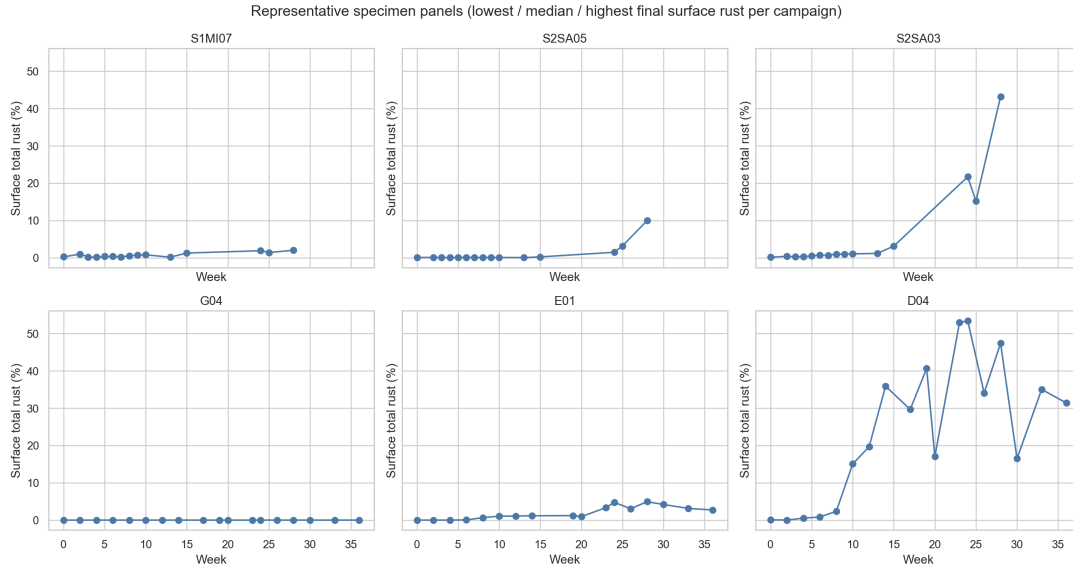


Figure 5: Representative specimen-level surface corrosion trajectories.

Interpretation and evidence strength. Figure 5 is strong evidence because it makes the longitudinal nature of the dataset concrete. It also shows that specimen trajectories are informative but not perfectly smooth, which is important when later discussing degradation smoothing.

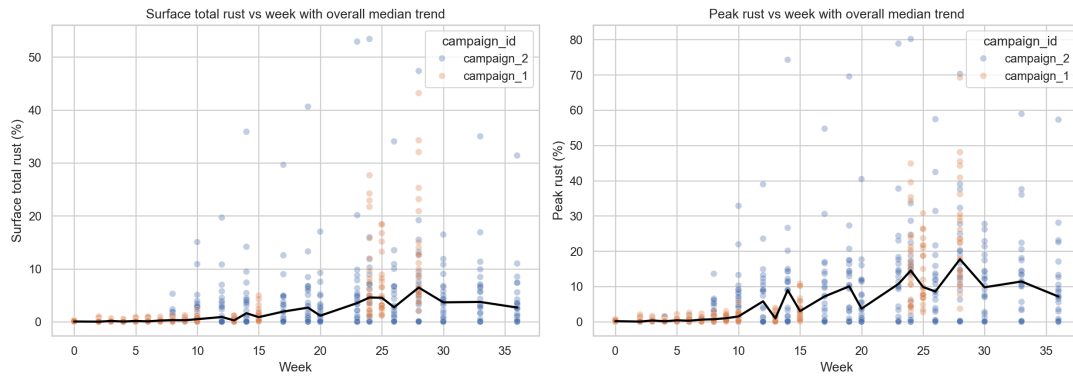


Figure 6: Global scatter-and-trend view of visible corrosion progression over time.

Interpretation and evidence strength. Figure 6 is strong evidence for the existence of a broad upward corrosion trend. It is especially useful when a presentation needs a single figure that summarizes progression without focusing on individual specimens.

6.6 Feature correlations

Figure 7 shows the focused feature-target correlation heatmap. This is one of the most important figures in the entire project because it clarifies what the engineered features actually capture.

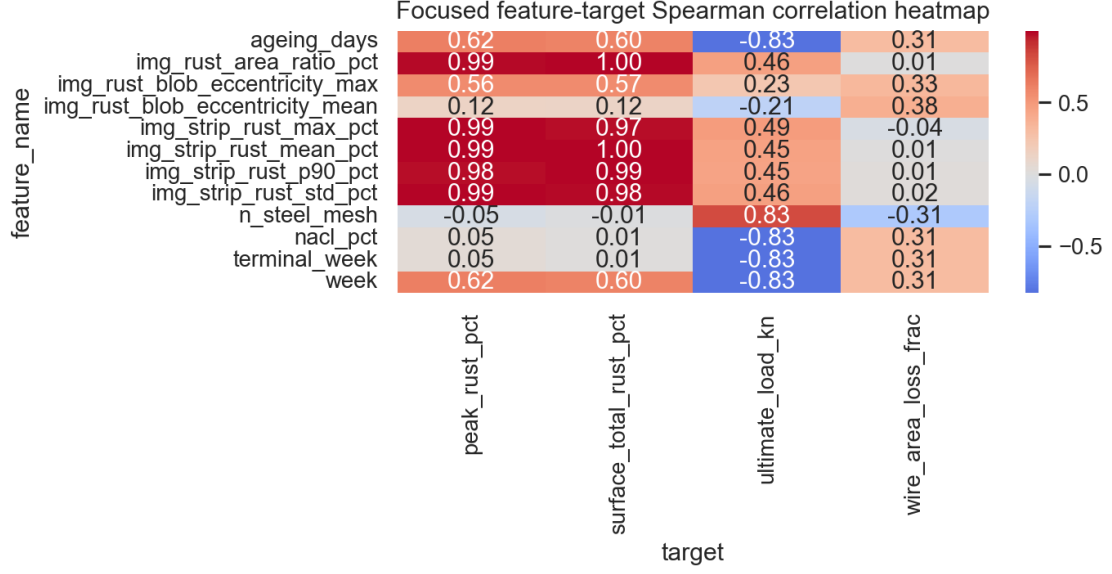


Figure 7: Focused feature-target correlation heatmap for visible corrosion targets and sparse structural targets.

Interpretation and evidence strength. Figure 7 is strong evidence. It shows that the engineered image features track visible corrosion very strongly, especially for `surface_total_rust_pct` and `peak_rust_pct`, but that the relationship to sparse structural targets is much weaker. The strongest univariate correlation with `wire_area_loss_frac` is only about 0.38, which already signals a hard hidden-damage problem before model fitting begins.

6.7 Redundancy and collinearity

Two complementary figures explain redundancy. Figure 8 shows the correlation structure among key image features. Figure 9 shows the highest-collinearity feature pairs that motivated explicit cleanup.

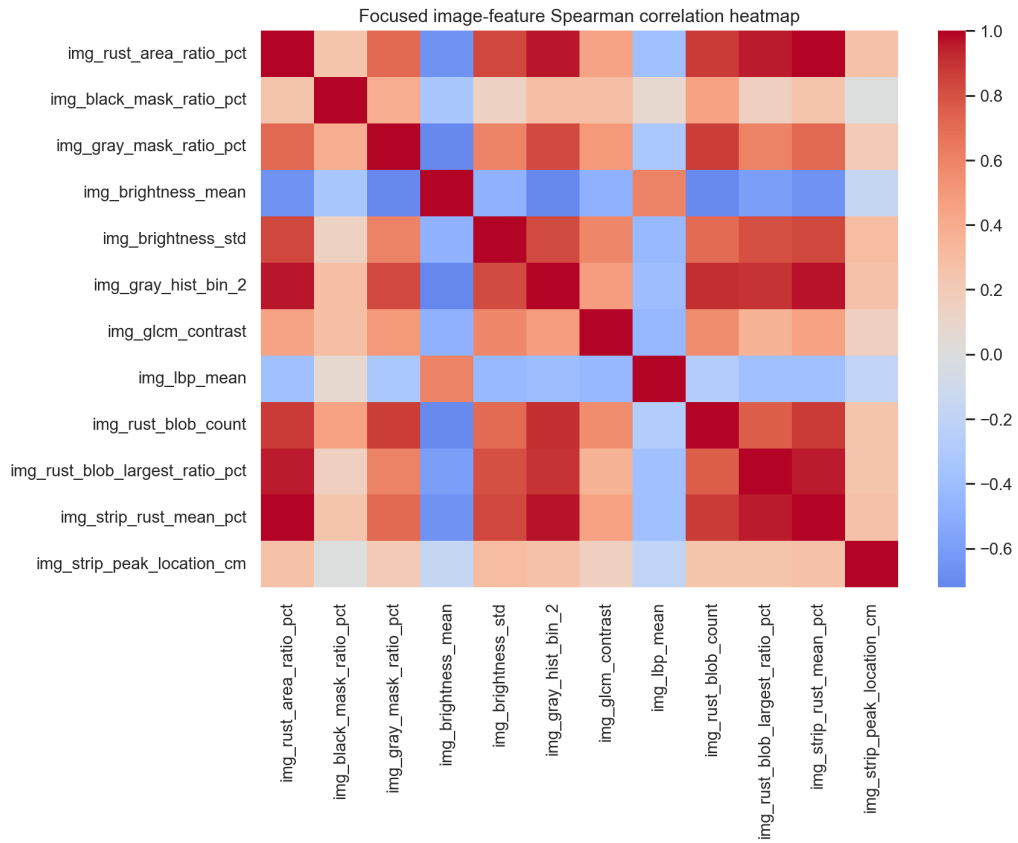


Figure 8: Correlation heatmap for key image features. Many features move together because they describe related aspects of visible corrosion.

Interpretation and evidence strength. Figure 8 is strong evidence that the feature space is not a collection of independent signals. Several image-derived features reflect the same underlying corrosion pattern, which helps explain why feature cleanup and family-based interpretation are necessary.

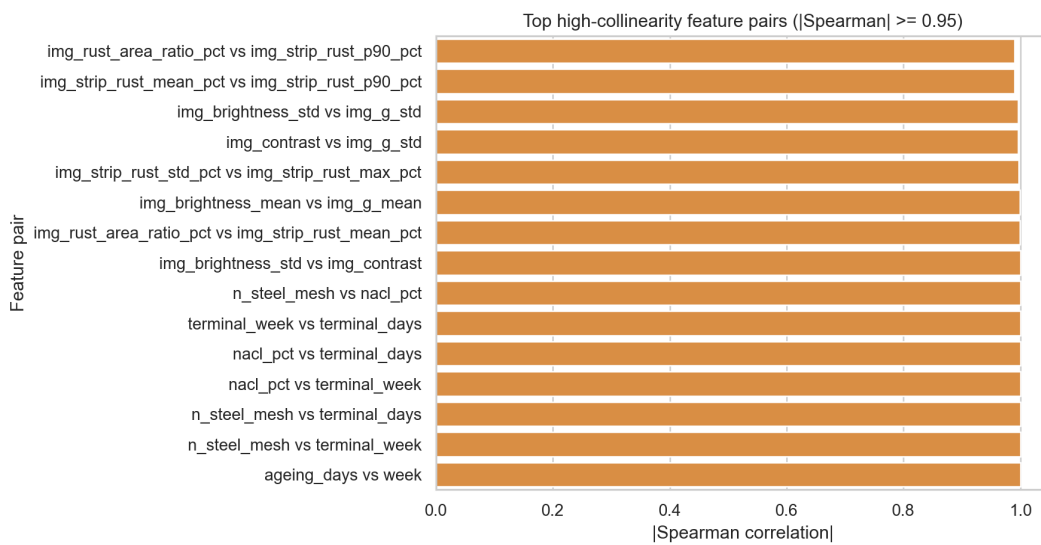


Figure 9: Highest-collinearity feature pairs found during diagnostics.

Interpretation and evidence strength. Figure 9 is strong evidence that redundancy is a

real modelling issue, not a cosmetic concern. The diagnostics found many highly collinear pairs, in addition to exact duplicates and near-constant features. This makes the later feature-space cleanup scientifically well motivated.

7 Model Benchmarking and Results

7.1 Grouped split results

The grouped split results provide the baseline within-dataset evaluation. The main grouped findings are:

- the surface stage performs strongly, but its main target is partly tautological,
- `peak_rust_pct` remains a useful auxiliary surface benchmark,
- the hidden-damage stage is much weaker, especially for `wire_area_loss_frac`.

Two grouped facts are particularly important:

- the extracted rust-area feature reconstructs `surface_total_rust_pct` almost exactly, with MAE 0.000228 and maximum absolute difference below 0.000500,
- the selected final hidden-damage models use only 8 metadata features for each structural target.

7.2 Leave-one-treatment-out

Leave-one-treatment-out is a harder test than grouped holdout because it asks whether a learned pattern transfers across treatment groups. For the current selected structural models:

- `wire_area_loss_frac`: MAE 0.116 and Spearman 0.186,
- `ultimate_load_kn`: MAE 0.203 and Spearman 0.122.

These values are not strong enough to claim reliable structural generalization, but they are still more informative than the grouped split alone.

7.3 Leave-one-campaign-out

Leave-one-campaign-out is the most scientifically important evaluation regime because it tests a real shift in experimental setting. It is also where the structural stage remains weakest.

For the final robustness-selected structural models:

- `wire_area_loss_frac`: MAE 0.121, Spearman -0.266,
- `ultimate_load_kn`: MAE 0.564, Spearman 0.353.

These results show that the project still does not support strong cross-campaign hidden-damage claims.

7.4 Robustness discussion

Benchmark robustness matters more than a single best grouped score because the grouped split is the easiest regime. Figure 10 summarizes how error changes when the split becomes harder.

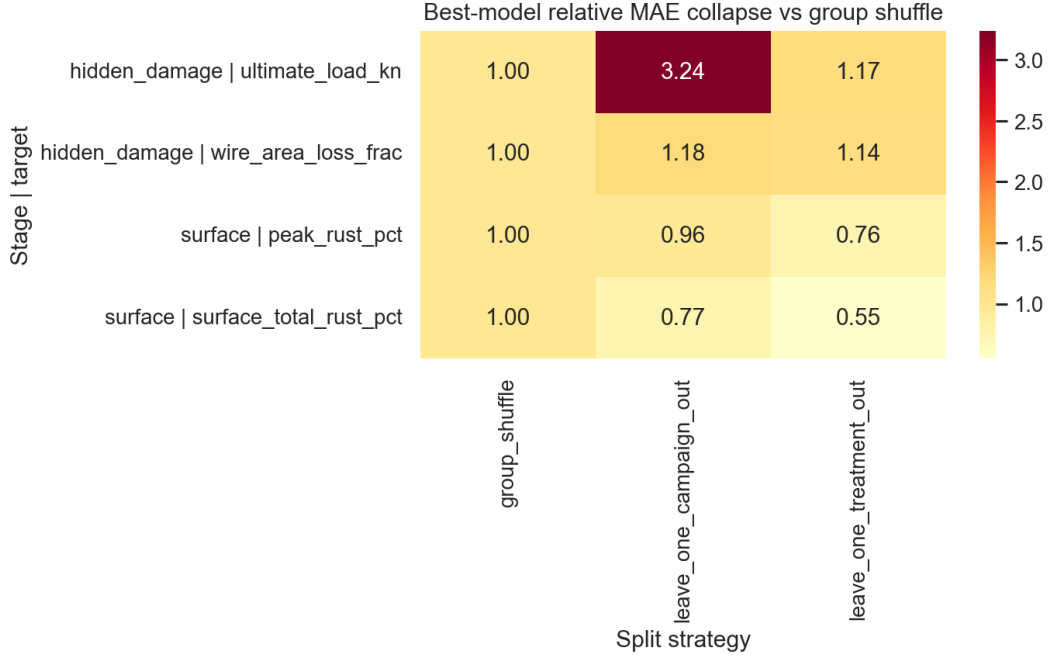


Figure 10: Relative MAE collapse across split strategies. Values above 1 indicate worse performance than the grouped baseline.

Interpretation and evidence strength. Figure 10 is strong evidence because it summarizes the central robustness lesson of the project. Surface benchmarks remain relatively stable, while structural benchmarks deteriorate markedly under harder evaluation.

Figure 11 complements that view by showing the structural best-model robustness directly.

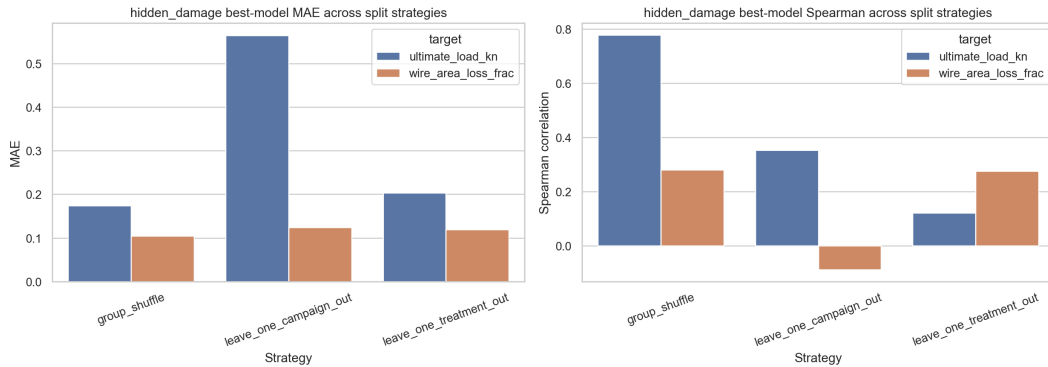


Figure 11: Robustness profile of the selected structural models across grouped, leave-one-treatment-out, and leave-one-campaign-out evaluation.

Interpretation and evidence strength. Figure 11 is strong evidence. It shows clearly that structural modelling is the weak part of the pipeline, and it also supports the conclusion that benchmark robustness is a better guide than one good average score.

Table 2 summarizes the current benchmark snapshot after feature cleanup and robustness-aware

model selection.

Table 2: Current benchmark snapshot for the most relevant targets. The structural rows correspond to the final robustness-selected models.

Target	Current model / feature set	Key metrics	Interpretation
surface_total_rust_pct	rustarea reconstruction check	MAE 0.000228, Spearman 0.9996 between label and extracted rust area	Strong as pipeline validation, but not a substantive hidden-damage result.
peak_rust_pct	Random Forest / full surface feature table	Grouped MAE 1.097, grouped Spearman 0.992, LOCO MAE 1.058, LOCO Spearman 0.977	Meaningful auxiliary visible-corrosion benchmark.
wire_area_loss_frac	CatBoost / metadata-only	Grouped MAE 0.124, grouped Spearman 0.289, LOCO MAE 0.121, LOCO Spearman -0.266	Weak and unstable. Better viewed as a difficult negative result than as a predictive success.
ultimate_load_kn	Random Forest / metadata-only	Grouped MAE 0.174, grouped Spearman 0.778, LOCO MAE 0.564, LOCO Spearman 0.353	Improved relative to the earlier baseline, but still too fragile for strong cross-campaign claims.

The improvement round is still important to mention. Compared with the earlier baseline, leave-one-campaign-out absolute error improved for both structural targets, but the improvement was uneven:

- `ultimate_load_kn` improved in a relatively clean way,
- `wire_area_loss_frac` improved in MAE on the harder splits but remained weak in ranking and even worsened in leave-one-campaign-out Spearman.

7.5 Degradation modelling

The degradation stage should be interpreted conservatively. It smooths model-generated hidden-damage proxies over time. Figure 12 shows why this smoothing was introduced.

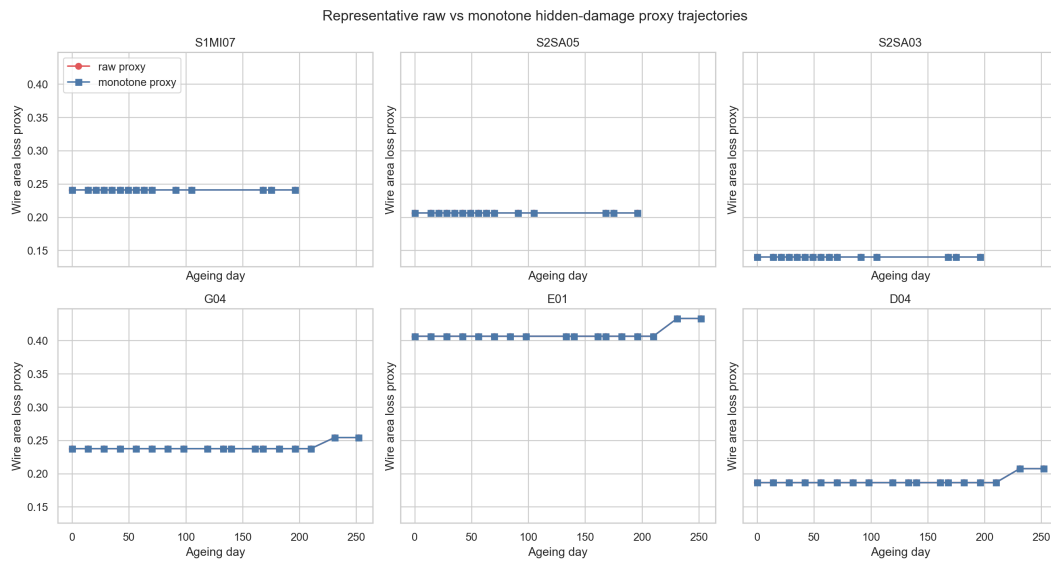


Figure 12: Representative raw hidden-damage proxy trajectories compared with their monotone-smoothed versions.

Interpretation and evidence strength. Figure 12 is moderate evidence. It honestly shows that the degradation stage is primarily a stabilizing descriptive layer, added because the raw hidden-damage proxy is not cleanly monotone over time.

7.6 Proxy-RUL logic

The final proxy-RUL output is best interpreted as threshold-status analysis. Figure 13 summarizes the current status.

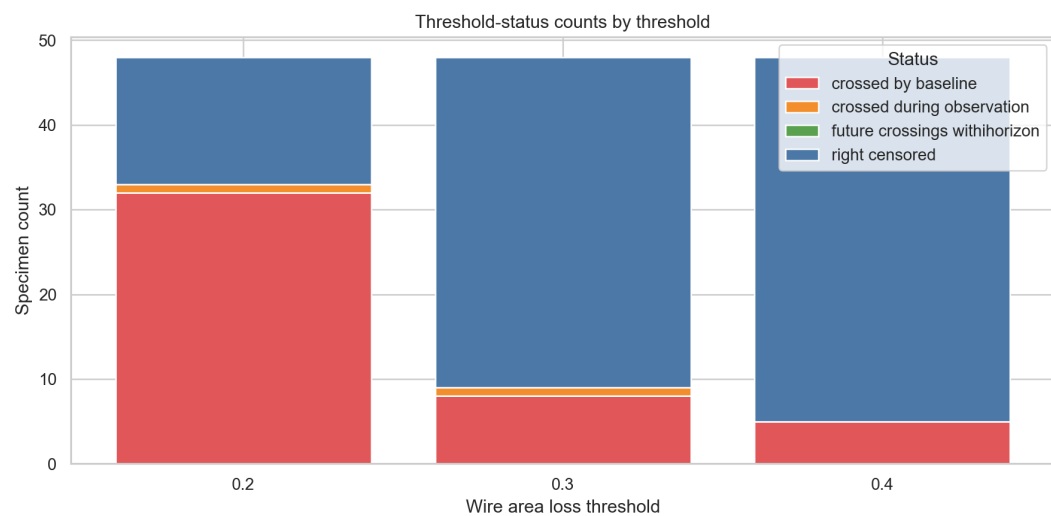


Figure 13: Threshold-status counts for the proxy-RUL stage. Most cases are already crossed or right-censored, and no future threshold crossings occur within the projection horizon.

Interpretation and evidence strength. Figure 13 is strong as an honesty figure, but weak as predictive evidence. It matters because it makes the current state of the proxy-RUL stage very clear: the pipeline does not yet provide useful forward residual-life estimates.

8 Important Figures and Their Interpretation

This section compares the most meaningful figures and explains why they are more useful than some visually attractive alternatives.

8.1 Figures that explain the dataset constraints

Figures 1 and 2 are the best pair for explaining the dataset limitation. The first figure shows the broad missingness structure; the second figure turns that into a direct structural-label message. Together they justify the entire staged formulation.

These figures are stronger than simple count charts such as row counts, campaign counts, or treatment counts because they explain not just *how much* data exists, but *what kind* of supervision is actually available.

8.2 Figures that explain corrosion progression

Figures 4, 5, and 6 work well as a sequence:

- Figure 4 shows the campaign-level trend,
- Figure 5 shows representative specimen-level behaviour,
- Figure 6 shows the broad global relationship with time.

Together they make the longitudinal corrosion story easy to explain in a presentation.

8.3 Figures that explain what the features learned

Figures 7, 8, and 9 are the strongest feature-interpretation figures:

- Figure 7 shows which feature families track visible corrosion well and how weak the structural links remain.
- Figure 8 shows that many image features measure related aspects of the same visible pattern.
- Figure 9 makes the redundancy problem concrete.

This set is more informative than raw feature-importance plots for the surface stage, because the surface stage is partly tautological and can otherwise be over-interpreted.

8.4 Figures that explain model robustness

Figures 10 and 11 are the best model-performance figures because they focus on robustness rather than just a single best number. They also communicate the most important scientific lesson from benchmarking:

Good grouped performance is not enough. The harder split regimes reveal whether the learned pattern is genuinely transferable.

8.5 Figures that explain why the final stages remain exploratory

Figures 12 and 13 are useful because they make the weakness of the final stages clear without making them look more mature than they are.

The degradation figure explains why smoothing was introduced. The proxy figure explains why the final stage remains exploratory. This is precisely why they should be shown, but shown carefully.

9 Limitations

The limitations are central to the integrity of the report and should be stated directly.

- **Sparse structural supervision.** Only 48 of 791 usable rows carry structural labels, and those labels occur only at terminal time points.
- **Surface-stage tautology risk.** The strongest visible target, `surface_total_rust_pct`, is almost numerically identical to the extracted rust-area feature, so this stage is primarily a validation of feature extraction rather than a major prediction result.
- **Campaign confounding.** Structural performance changes sharply under leave-one-campaign-out evaluation, indicating that the learned signal is not yet robust across campaigns.
- **Metadata-driven hidden-damage improvement.** The improvement round selected metadata-only models for both structural targets, which is informative but also shows that image features are not yet delivering robust structural generalization.
- **Exploratory degradation and proxy-RUL stages.** The degradation stage smooths model-generated hidden-damage proxies, and the proxy stage currently yields threshold-status summaries rather than useful forward residual-life estimates.
- **Incomplete recovery of the original image-preprocessing pipeline.** The project uses the provided PNG images directly because the exact original GIMP workflow could not be fully reconstructed from the available documentation.

10 Conclusion

The project has successfully achieved a strong and presentation-worthy baseline in the following sense:

- the data were audited carefully and aligned into a trustworthy canonical dataset,
- campaign and treatment mapping were made deterministic and reviewable,
- interpretable image features were extracted systematically,
- temporal corrosion progression was documented clearly,
- diagnostics revealed real feature redundancy and justified feature cleanup,

- benchmark robustness was treated as the main evaluation principle rather than a single best score.

The strongest conclusions are not only positive ones. The project also produced valuable negative findings:

- hidden-damage generalization remains weak,
- cross-campaign structural prediction is not yet reliable,
- proxy-RUL remains exploratory.

That combination — strong audit and diagnostics, interpretable feature engineering, meaningful temporal analysis, and honest benchmark limitations — makes the current repository scientifically useful even though the final structural and proxy stages remain limited.

11 Next Steps

The next scientifically defensible step should stay within the current conservative philosophy. The priority is not to add a new flashy model family, but to reduce shortcut dependence and clarify how much structural signal remains once campaign-sensitive metadata are restricted more aggressively.

The most useful next actions are:

- continue analysing feature subsets that separate image-derived information from campaign-linked metadata,
- strengthen figure-based comparisons of visible progression and structural weakness,
- preserve the current grouped and leave-one-campaign-out evaluation framework as the main robustness test,
- treat degradation and proxy-RUL as exploratory until stronger structural supervision becomes available.

In presentation terms, the strongest story is therefore:

The project built a trustworthy, interpretable, leakage-safe baseline and made the scientific limits of the current dataset visible. It is strongest in data validation, visible corrosion analysis, temporal progression, feature diagnostics, and robustness analysis. It is not yet strong enough to claim robust hidden-damage prognostics or decision-ready proxy-RUL.